

Global Renewable Energy in 2025: How Far Have We Come?

A data-driven overview of renewable energy's growing share in the global energy and electricity mix, spanning hydropower, wind, and solar technologies.

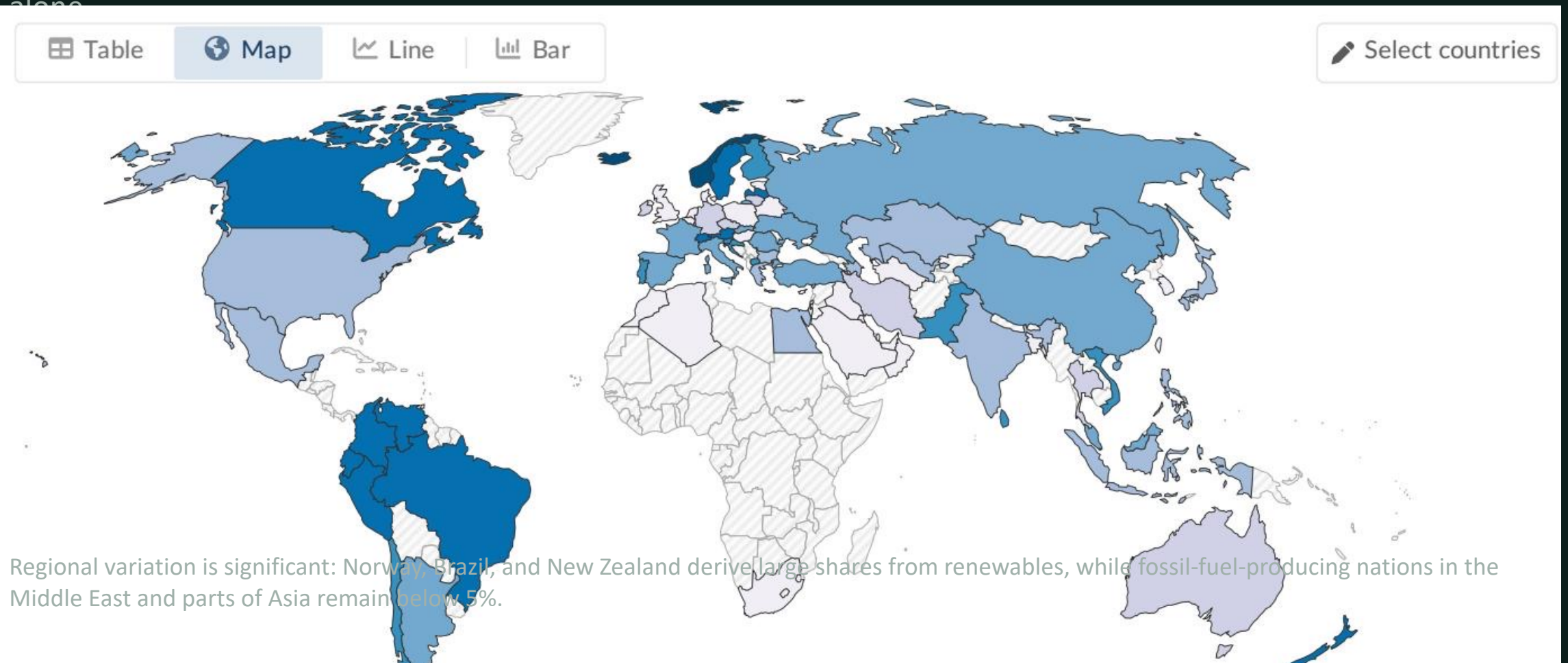
Source: Our World in Data – Renewable Energy (Ember 2026; Energy Institute 2025)

75% of Global GHG Emissions Still Come from Fossil Fuels

- Since the Industrial Revolution, fossil fuels have dominated the global energy mix.
- Burning fossil fuels for energy accounts for three-quarters of all greenhouse-gas emissions.
- Fossil-fuel combustion causes at least 5 million premature deaths annually from local air pollution.
- A rapid shift toward low-carbon sources — nuclear and renewable technologies — is essential to cut CO₂ and improve air quality.
- Renewable energy includes solar, wind, hydropower, geothermal, bioenergy, wave, and tidal sources.

Renewables Now Supply ~14% of Global Primary Energy

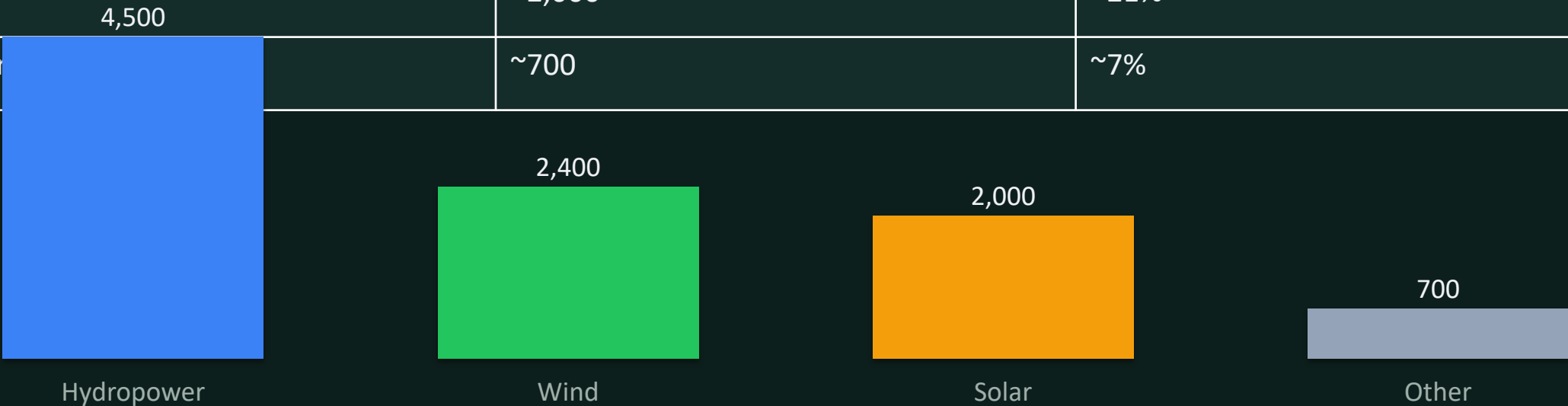
Approximately one-seventh of the world's primary energy now comes from renewable technologies (2024 data, substitution method). Primary energy encompasses electricity, transport, and heating combined. Because transport and heating are harder to decarbonize, the renewable share in total energy is lower than in electricity alone.



Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar

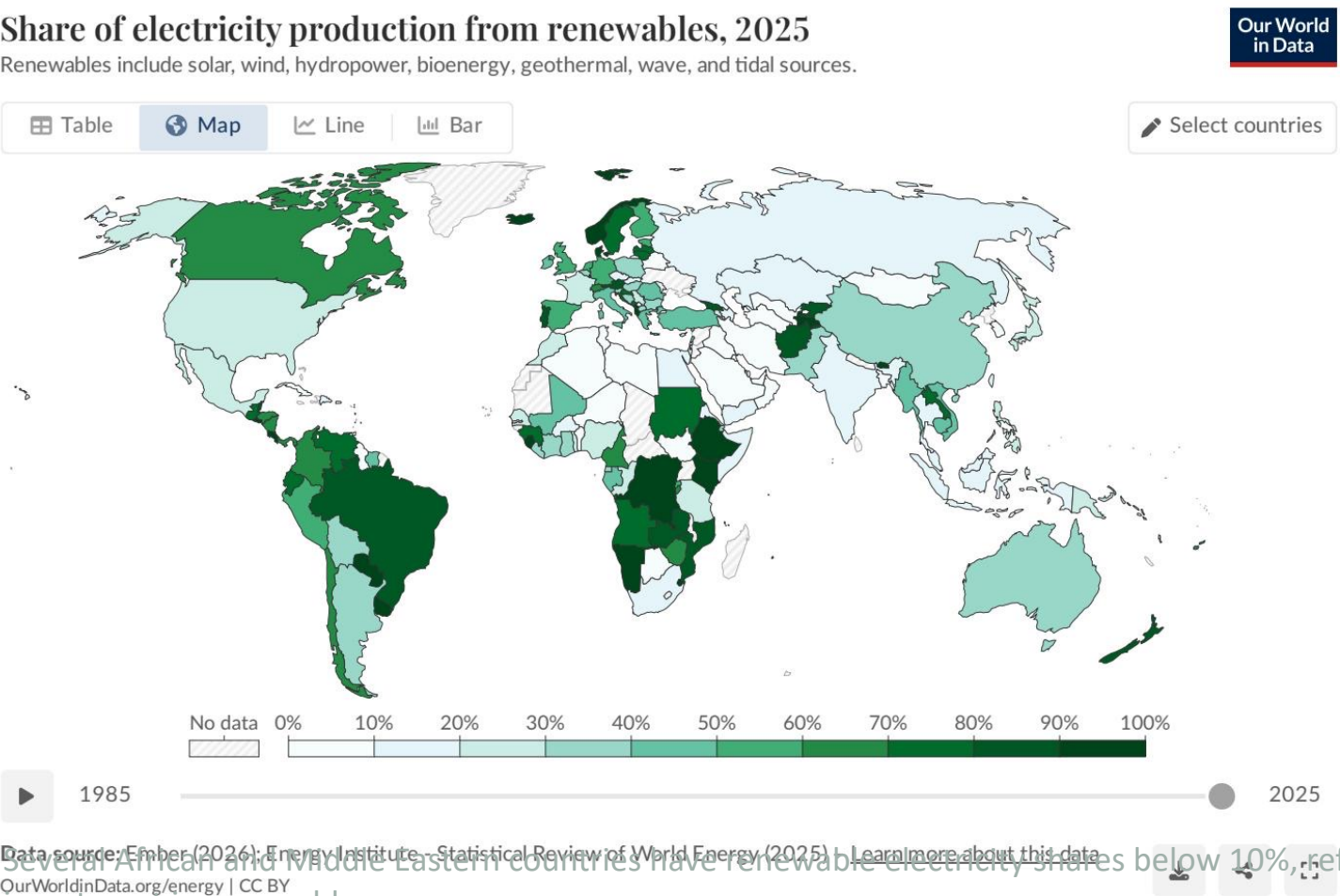
Wind and solar combined now nearly match hydropower's output and are growing much faster.

Technology	Approx. Global Generation (2024, TWh)	Share of Renewable Electricity
Hydropower	~4,500	~47%
Wind	~2,400	~25%
Solar	~2,000	~21%
Other	~700	~7%



Almost One-Third of Global Electricity Now Comes from Renewables

Globally, almost one-third of electricity now comes from renewable sources (2025). This is substantially higher than the primary energy share because electricity is the sector where renewables have made the most progress.

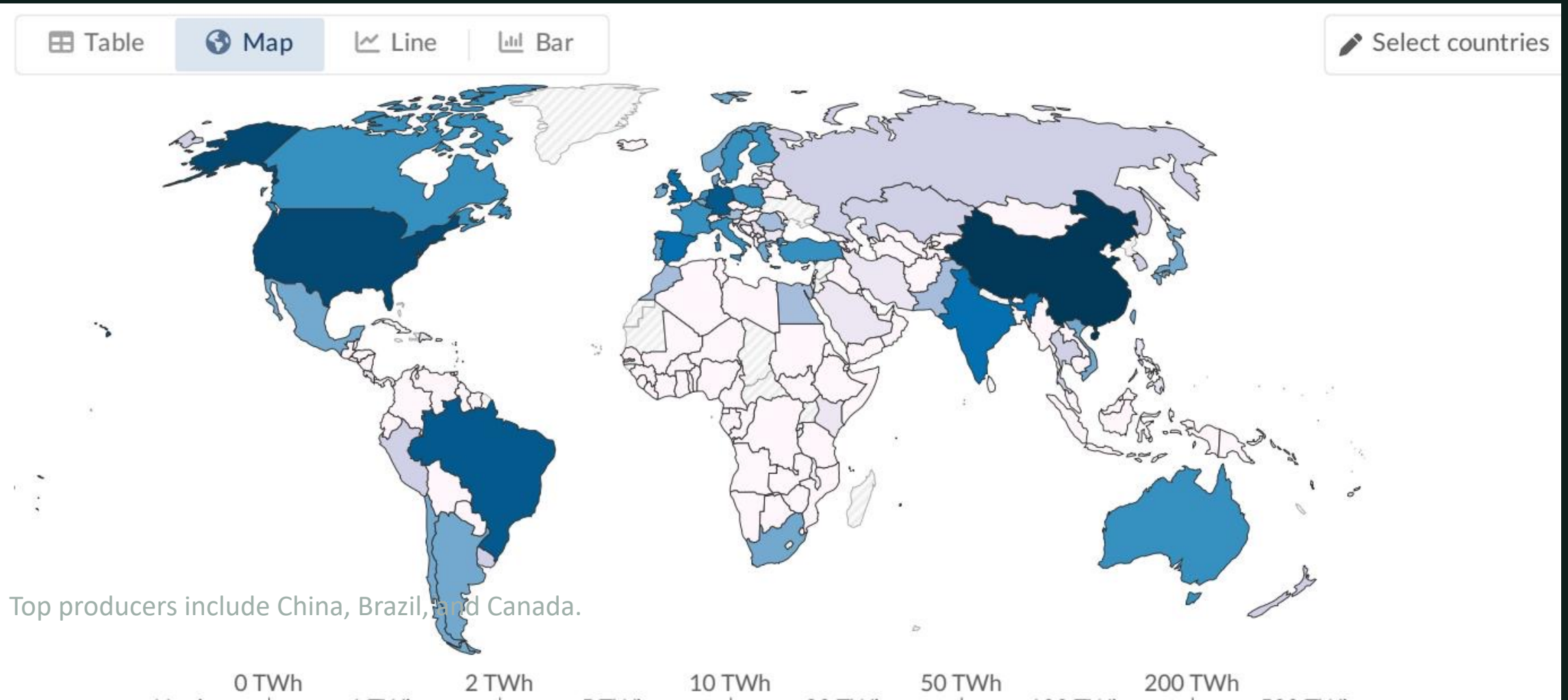


Region / Country	Renewable Share of Electricity
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%

Several African and Middle Eastern countries have renewable electricity shares below 10%, reflecting both fossil fuel dependence and limited investment in renewables.

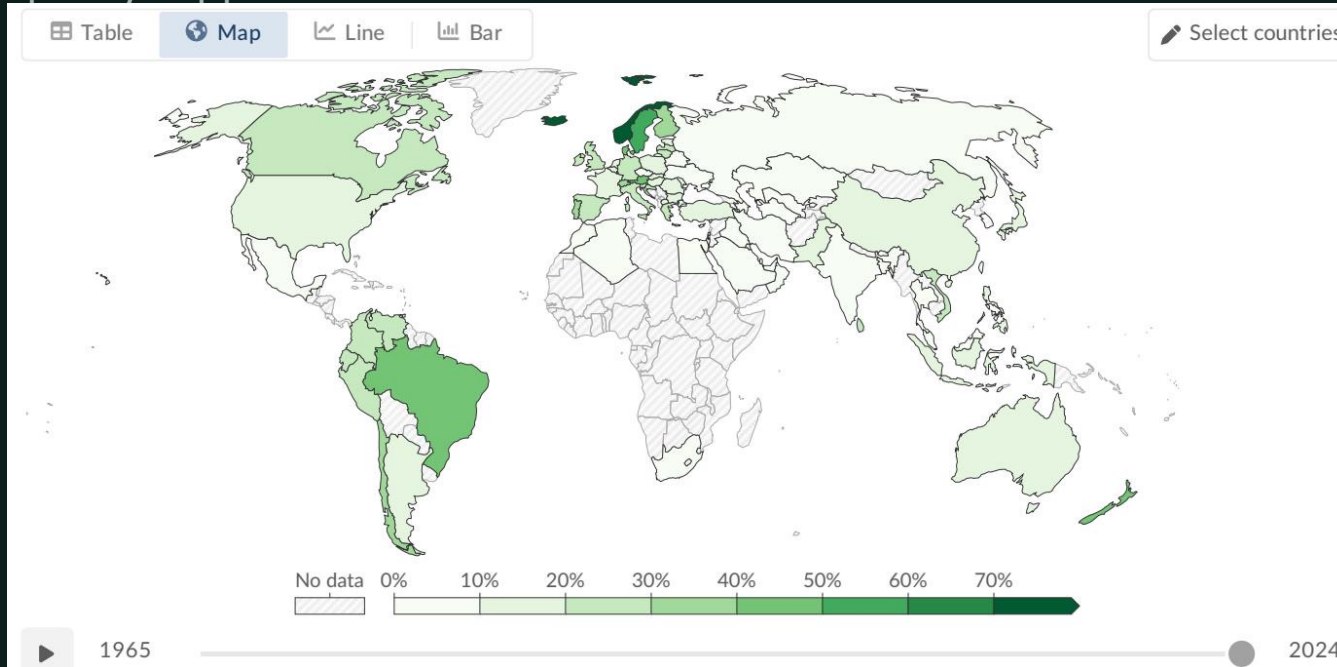
Hydropower Remains the Largest Renewable Source at ~50% of Generation

Hydropower has been generating electricity at scale for over a century and remains the single largest renewable electricity source, accounting for roughly half of all renewable generation.



Wind and Solar Are the Fastest-Growing Energy Technologies in History

Solar is the fastest-growing source, with dramatic global generation increases since 2010 driven by falling costs and policy support.

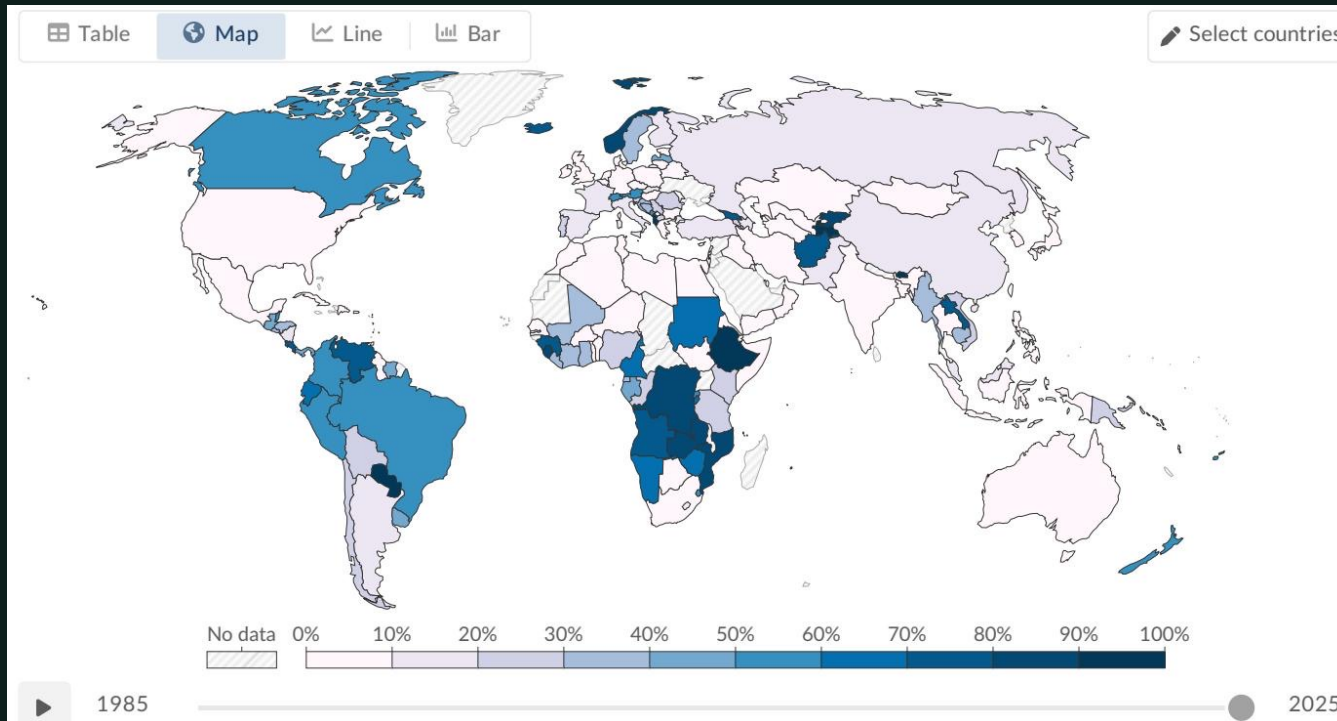


Wind vs. Solar (2024)

- Wind: ~2,400 TWh (~25% of renewable electricity)
- Solar: ~2,000 TWh (~21% of renewable electricity)
- Wind energy has grown rapidly and is now a major contributor, especially in Europe, China, and the United States.
- Solar generation has increased dramatically since 2010.
- Combined, wind + solar now nearly match hydropower output.

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin America Lead

The transition to renewables is highly uneven across regions.



Regional Snapshot

- Scandinavia, Brazil, Canada: 65–98% renewable electricity
- Denmark & Germany: heavy wind and solar investment
- Sub-Saharan Africa: minimal renewable electricity infrastructure
- Middle East: many nations remain almost entirely fossil-fuel dependent
- Several African and Middle Eastern countries: renewable electricity shares below 10%

Key Takeaways: Renewables Are Growing Fast but the Transition Must Accelerate

- ~14% of global primary energy and ~30% of global electricity now come from renewable sources — a significant milestone but far from sufficient for climate targets.
- Hydropower remains dominant, generating roughly half of all renewable electricity, but its growth potential is limited compared to wind and solar.
- Wind and solar are the fastest-growing energy technologies in history; combined, they now produce nearly as much electricity as hydropower.
- Stark regional disparities persist: Scandinavia and Latin America exceed 60–90% renewable electricity, while much of the Middle East and Sub-Saharan Africa remain below 10%.
- Transport and heating lag far behind electricity in the renewable transition, keeping the overall primary energy share much lower than the electricity share.

Accelerating deployment — especially in wind, solar, and grid infrastructure — is essential to close the gap between today's progress and climate-safe energy systems.